A396102
Congruences for an inverse-iterate family

Dev Goyal

24 October 2026

AI contribution and review status. ChatGPT (OpenAI) generated the mathematical exposition, proof development, Lean interfaces, and checks for this collection in response to Dev Goyal's prompts. Dev Goyal is the sole human author. Independent human review is pending. Original sequence authors, known mathematical results, and adapted upstream code retain their separate credits. This document makes no claim of publication, OEIS acceptance, or worldwide priority.

Abstract

We construct the unique normalized integer solution of $A(A(A(x))) = (1+x)A(A(x))$ and prove that every positive-index coefficient is one modulo three. A parameter theorem determines every natural compositional iterate modulo the parameter, with an explicit source-equation and inverse bridge.

1 A396102 and the inverse-iterate family

For each positive integer r, consider the normalized series equation

$$F_r^{\circ r}(x) = (1+x)F_r^{\circ(r-1)}(x). \tag{1}$$

The case $r = 3$ is the original A396102 equation. We construct it through an auxiliary series, rather than assuming that a solution exists.

Lemma 1.1. *For every $r \geq 0$ and every commutative coefficient ring, there is a unique normalized series R_r satisfying*

$$R_r = x - R_r R_r^{\circ r}. \tag{2}$$

If $r \geq 1$, its compositional inverse F_r is the unique normalized solution of (1).

Proof. Define $T_r(G) = x - GG^{\circ r}$. It preserves normalization. If two normalized series agree through degree $n-1$, their iterates do also. The difference of the two products $GG^{\circ r}$ and $HH^{\circ r}$ has one factor divisible by x^n and another divisible by x in each term. Thus

$$G - H = O(x^n) \quad \Longrightarrow \quad T_r(G) - T_r(H) = O(x^{n+1}).$$

Starting at x and iterating T_r, every coefficient stabilizes after finitely many steps. The stabilized coefficients give a fixed point; the same order-increase implication proves uniqueness. This is a purely coefficientwise construction and works over arbitrary commutative rings.

Every normalized series U has a unique normalized compositional inverse. Indeed, in $U(V) = x$ the coefficient of degree $n \geq 2$ is v_n plus an expression in earlier coefficients of V, so it determines v_n with multiplier one and no division. This constructs a right inverse V. Construct

a right inverse W of V in the same way; associativity gives $U = U \circ (V \circ W) = (U \circ V) \circ W = W$, hence $V \circ U = x$ as well. The coefficient recursion proves uniqueness.

Take the genuine compositional inverse F_r of R_r. Substitute $F_r^{\circ r}(x)$ into (2). Inverse cancellation gives

$$F_r^{\circ(r-1)} = F_r^{\circ r} - xF_r^{\circ(r-1)},$$

which is (1). Conversely, substitute $R^{\circ r}$ into (1), where $R = F^{-1}$. This gives $x = (1 + R^{\circ r})R$, hence (2). Uniqueness of R_r now implies uniqueness of F_r. □

Theorem 1.2. *For every $r \geq 1$,*

$$R_r(x) \equiv \frac{x}{1+x} \pmod{r}, \qquad F_r(x) \equiv \frac{x}{1-x} \pmod{r}.$$

Consequently, for $n \geq 1$ and $k \geq 0$,

$$[x^n]F_r^{\circ k}(x) \equiv k^{n-1} \pmod{r}. \tag{3}$$

Here $0^0 = 1$, consistently with the degree-one coefficient of the zeroth compositional iterate x.

Proof. Write $M_c(x) = x/(1 - cx)$. Direct substitution gives $M_a \circ M_b = M_{a+b}$ over any commutative ring. Therefore $M_{-1}^{\circ r} = M_{-r} = x$ over $\mathbb{Z}/r\mathbb{Z}$. Moreover

$$x - M_{-1}(x)x = M_{-1}(x),$$

so M_{-1} satisfies the auxiliary fixed-point equation modulo r. The coefficientwise uniqueness in Lemma 1.1 identifies it with the reduction of R_r. Its actual inverse is M_1, identifying the reduction of F_r. Finally, $M_1^{\circ k} = M_k = x/(1 - kx)$, whose degree-$n$ coefficient is k^{n-1} for $n \geq 1$. □

For $r = 3$ and $k = 1$, this proves the A396102 conjecture

$$[x^n]F_3(x) \equiv 1 \pmod{3} \qquad (n \geq 1).$$

The formal proof includes both directions of the correspondence with the original triple-composition equation, not merely a congruence for an auxiliary series. The formal parameterization uses $r + 1$ when stating the original equation so that its predecessor iterate is always a natural number. The coefficient theorem itself also has a well-defined $r = 0$ version in `ZMod 0`; no such extension is needed to settle the stated OEIS problem.

2 Lean proof and exact scope

The dedicated entry file is `lean/Entries/A396102.lean`. Its declarations lie in namespace `FinalCollection.A396102`. The principal checked endpoints are:

- `source_equation`: the exact triple-composition equation.
- `source_existence_uniqueness`: existence and uniqueness of its normalized solution.
- `coefficient_congruence`: the source assertion for every $n \geq 1$.
- `iterate_congruence`: every natural iterate modulo three.
- `family_source_existence`: the entire family of original equations.
- `family_source_equation`: the canonical family solves that equation.
- `auxiliary_source_equation`: the fixed-point equation constructing the inverse.
- `inverse_right, inverse_left`: both genuine inverse identities.
- `family_coefficient_congruence`: every coefficient modulo the parameter.
- `family_iterate_congruence`: the general iterate formula.

The full implementations are in `lean/More/InverseIterate.lean`. The source offset is one. The formal original-family statement uses parameter $r + 1$, while $A396102 = F_3$. Thus $r = 2$ in the original-family existence theorem gives the required triple equation. The congruence statements also allow the formal convention `ZMod 0`, but the mathematical modulus assertions here use positive parameters.

The companion is a separate, explicitly typed Lean entry file. Its proofs invoke the preserved implementation modules; the complete dependency source is included in the same archive. Shared lemmas are reused rather than assumed. The entry file is not a self-contained replacement for Mathlib or the supplied common library.

The project pins Lean 4.19.0 and Mathlib revision

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

The collection checker recompiles the proof sources and audits the theorems' transitive axiom dependencies. Only `propext`, `Classical.choice`, and `Quot.sound` are permitted. The delivered verification reports record the actual successful runs. Finite numerical checks are corroboration; the displayed conclusions are unbounded theorems. An AI review of source correspondence does not replace independent human review.

References

[1] OEIS Foundation, entry A396102, `https://oeis.org/A396102`. Source checked 24 October 2026. The source sequence and selected conjectures are credited to Paul D. Hanna.

[2] D. Goyal, supplied reference repository, `https://github.com/dvgyl/oeis-a396843-proof`. This separate presentation belongs to the final collected proof package; no earlier DOI is reassigned to it.